Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). 12 local configuration/tokenizer/data checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status

GPU production queued after eleven preemptions; step-3848 checkpoint saved. Measured speed: 5.6 seconds/update, excluding interruptions. TPU trial submitted on 2026-10-14: v6e-32, 32 chips, us-east5-b. Twenty updates on existing regional native caches; same model/batch shape. TPU Splash attention, fresh initialization, compressed smoke schedule. Production migration requires 320.4 GB of MPNN caches and optimizer state. No bulk transfer performed; explicit cross-region transfer approval required. Contact-accuracy results are pending.